Zone-Market Clearing: a dimension-free price search for pure equilibria of aggregative carrier games with a routing oracle

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Theory notes, direction A

Abstract

We study the affine, exposure-based (D1) carrier game of `theory/main.tex`. N carriers choose plans from huge finite sets through an NP-hard routing oracle, and they interact only through $d = 2Z$ zone aggregates. Each oracle call is one routing solve. We propose *Zone-Market Clearing* (ZMC). All carriers are priced at one common vector $\lambda \in \mathbb{R}^d$, the shadow price of the zone aggregates in the dual of the convexified potential. Each round costs N independent (parallel) routing solves and returns, for *every* zone at once, the signed clearing residual $s_k = A_k(\lambda) - \lambda_k/M_k$. Each coordinate keeps its own bracket and moves by safeguarded secant steps. Two structural results support the method. First, a *clearing identity* (Theorem 2): for any price and any profile of pricing-optimal plans, the potential gap to the convexified optimum is exactly $\frac{1}{2}\|e\|_M^2$, where $e = M^{-1}\lambda - A$ is the market-clearing error, and each carrier's Nash regret is at most $\frac{1}{2}\|e\|_M^2$ restricted to its own zones. The own-contribution term, which appears in the Shapley–Folkman bounds and cannot be removed there, cancels exactly. Second, a *diagonal monotonicity* theorem (Theorem 6): the coordinatewise sign of the residual localises the optimal price coordinate up to $M_k\kappa_k$, where κ_k is the cross-zone variation of the response. Hence T rounds, that is NT oracle calls, give every coordinate to accuracy $M_k A_k^{\max} 2^{-\lceil T/2 \rceil} + M_k\kappa_k$, independently of Z. Black-box methods on the same dual pay either $1/\varepsilon$ calls (Frank–Wolfe, subgradient) or a factor d (cutting planes). The pure equilibrium is then obtained by oracle-free rounding over the collected plans and a best-response polish whose length is bounded by the clearing error. In prototypes ($Z \le 250$, $N \le 6$) ZMC needs a constant number of rounds as Z grows. At an equal budget of about 60 oracle calls, its pure equilibria have a smaller potential gap than those of Frank–Wolfe and column generation in 17 of 20 instances. They come within 20% of full column generation, which needs 3 to 13 times more calls. On nine real city maps with a zone-level proxy oracle, the advantage is clear only on the largest map (Berlin, 593 zones). On the others all methods, plain best response included, are within a few percent of each other. Section 7 lists the limits: the floor $M_k\kappa_k$, the restriction to D1 (and to Γ-maximin through the surcharge), and the proxy oracle used for the city maps.

1 Setting

Assumption 1 (D1 affine aggregative game)**.** There are N carriers and d aggregate coordinates. Carrier i has a finite plan set $\mathcal{P}_i$. Each plan $p \in \mathcal{P}_i$ has a contribution $a_i(p) \in \mathbb{R}^d_{\ge 0}$ and a private cost $c_i(p) \in \mathbb{R}$; the private cost includes the carrier's own-term and any charges $\langle t, a_i(p)\rangle$ posted by the city. The coupling matrix is $M = \mathrm{diag}(M_1, \dots, M_d)$ with $M_k > 0$. At a pure profile σ, with

$A(\sigma) = \sum_i a_i(\sigma_i)$ and $A_{-i} = A - a_i(\sigma_i)$, carrier i pays

$$J_i(p; \sigma_{-i}) = c_i(p) + \langle a_i(p), MA_{-i}(\sigma)\rangle.$$

This is Example 2.2 of `main.tex` under D1 with costs instead of profits, with $a = (q, \tau)$ and $M = \mathrm{diag}(b, \gamma c)$. Coordinates with $M_k = 0$ carry no interaction and are dropped. By Lemma 3.6 of `main.tex`, Γ-maximin equilibria under credal boxes are the pure Nash equilibria of the same game with the surcharge $\delta\langle a_i(p), Mw\rangle + \gamma\langle t_i(p), \bar{\mu}\rangle$ added to c_i. Everything below therefore applies verbatim to the Γ-maximin problem.

Put $\tilde{c}_i(p) = c_i(p) - \frac{1}{2}\|a_i(p)\|_M^2$, where $\|v\|_M^2 = \langle v, Mv\rangle$ and $\|v\|_{M^{-1}}^2 = \langle v, M^{-1}v\rangle$. The function

$$\Phi(\sigma) = \sum_i \tilde{c}_i(\sigma_i) + \tfrac{1}{2}\|A(\sigma)\|_M^2$$

is an exact potential: $\Phi(p', \sigma_{-i}) - \Phi(\sigma) = J_i(p'; \sigma_{-i}) - J_i(\sigma_i; \sigma_{-i})$. This is Proposition 7.5 of `main.tex` with $\Phi = -\Psi$; the check is two lines, since $\frac{1}{2}\|R + a'\|_M^2 - \frac{1}{2}\|R + a\|_M^2 - \frac{1}{2}\|a'\|_M^2 + \frac{1}{2}\|a\|_M^2 = \langle a' - a, MR\rangle$.

Assumption 2 (Pricing oracle)**.** For every $\lambda \in \mathbb{R}^d$ and every i, an oracle returns some $p \in O_i(\lambda) := \arg\min_{p \in \mathcal{P}_i}\{\tilde{c}_i(p) + \langle \lambda, a_i(p)\rangle\}$.

This is the oracle $\mathcal{O}_i$ of `main.tex` (Assumption 2.5) with $Q = M$: a prize-collecting routing problem with zone prices and a separable quadratic per-zone term. The best-response oracle is the special case $\tilde{c}_i \to c_i$ and $\lambda = MA_{-i}$.

Value functions and the price dual. Let $V_i(\lambda) = \min_{p \in \mathcal{P}_i}\{\tilde{c}_i(p) + \langle \lambda, a_i(p)\rangle\}$, $V = \sum_i V_i$ and

$$G(\lambda) = V(\lambda) - \tfrac{1}{2}\|\lambda\|_{M^{-1}}^2.$$

V_i is a minimum of finitely many affine functions of λ, hence concave and piecewise linear. For $p \in O_i(\lambda)$ and all λ', $V_i(\lambda') \le \tilde{c}_i(p) + \langle \lambda', a_i(p)\rangle = V_i(\lambda) + \langle a_i(p), \lambda' - \lambda\rangle$. So $a_i(p) \in \partial^+ V_i(\lambda)$, the superdifferential. Write $R(\lambda) = \partial^+ V(\lambda)$. By the sum rule for finite concave functions on $\mathbb{R}^d$ (Rockafellar, *Convex Analysis*, Thm. 23.8), $R(\lambda) = \sum_i \partial^+ V_i(\lambda)$. In particular, for any oracle answers $\sigma_i \in O_i(\lambda)$, the observed aggregate $A(\sigma)$ lies in $R(\lambda)$.

The convexified potential is $\bar{\Phi}(x) = \sum_i \sum_p x_i(p)\tilde{c}_i(p) + \frac{1}{2}\|\bar{A}(x)\|_M^2$ on $X = \prod_i \Delta(\mathcal{P}_i)$, with $\bar{A}(x) = \sum_i \sum_p x_i(p) a_i(p)$. Let $\bar{\Phi}^* = \min_X \bar{\Phi}$ and $A_k^{\max} = \sum_i \max_{p \in \mathcal{P}_i} a_{ik}(p)$. Let $B = \prod_k [0, M_k A_k^{\max}]$.

Lemma 1 (Price duality)**.** *(a) G is strongly concave and has a unique maximiser λ^*; moreover* $\max G = \bar{\Phi}^* \le \min_\sigma \Phi(\sigma)$.

(b) All minimisers of $\bar{\Phi}$ have the same aggregate $\bar{A}^*$, and $\lambda^* = M\bar{A}^* \in B$.

(c) $0 \in R(\lambda^*) - M^{-1}\lambda^*$; that is, $M^{-1}\lambda^* \in R(\lambda^*)$.

Proof. (a) V is concave and finite, and $\frac{1}{2}\|\lambda\|_{M^{-1}}^2$ is strongly convex, so G is strongly concave and continuous. It is bounded above and coercive to $-\infty$ because V grows at most linearly. Hence a unique maximiser exists. Let $L(x, \lambda) = \sum_i \langle x_i, \tilde{c}_i\rangle + \langle \lambda, \bar{A}(x)\rangle - \frac{1}{2}\|\lambda\|_{M^{-1}}^2$. It is affine in x on the compact convex set X and concave and continuous in λ. By Sion's minimax theorem, $\min_x \sup_\lambda L = \sup_\lambda \min_x L$. For fixed x, $\sup_\lambda L$ is attained at $\lambda = M\bar{A}(x)$ with value $\bar{\Phi}(x)$. For fixed λ, $\min_x L = \sum_i V_i(\lambda) - \frac{1}{2}\|\lambda\|_{M^{-1}}^2 = G(\lambda)$, because a linear function on a product of simplices is minimised at a vertex, carrier by carrier. Hence $\bar{\Phi}^* = \max G$. Pure profiles are vertices of X, where $\bar{\Phi} = \Phi$, so $\bar{\Phi}^* \le \min \Phi$.

(b) If x, x' are minimisers with $\bar{A}(x) \neq \bar{A}(x')$, then $\bar{\Phi}((x+x')/2) < \frac{1}{2}\bar{\Phi}(x) + \frac{1}{2}\bar{\Phi}(x')$, because the linear part is affine and $\frac{1}{2}\|\cdot\|_M^2$ is strictly convex. This contradicts minimality. Let x^* be a minimiser. Since both sides of the minimax equality are attained, (x^*, λ^*) is a saddle point: $L(x^*, \lambda^*) \geq L(x^*, \lambda)$ for all λ. Indeed $\bar{\Phi}^* = \sup_\lambda L(x^*, \lambda) \geq L(x^*, \lambda^*) \geq \min_x L(x, \lambda^*) = G(\lambda^*) = \bar{\Phi}^*$. The unique maximiser of $L(x^*, \cdot)$ is $M\bar{A}(x^*)$, so $\lambda^* = M\bar{A}^*$. Finally $0 \leq \bar{A}_k^* \leq A_k^{\max}$ because each $\bar{A}_k$ is a sum over carriers of convex combinations of the numbers $a_{ik}(p) \in [0, \max_p a_{ik}(p)]$.

(c) This is the optimality condition $0 \in \partial^+ G(\lambda^*) = \partial^+ V(\lambda^*) - M^{-1}\lambda^*$. The sum rule applies because the quadratic is differentiable. □

2 The clearing identity

For a price λ and a pure profile σ, the *clearing error* is $e(\lambda, \sigma) = M^{-1}\lambda - A(\sigma)$, supply minus demand at price λ. Let $K_i = \{k : \ a_{ik}(p) \neq a_{ik}(p') \text{ for some } p, p' \in \mathcal{P}_i\}$ be the coordinates in which carrier i's plans differ (its zones). Write $\|v\|_{M,K}^2 = \sum_{k \in K} M_k v_k^2$.

Theorem 2 (Clearing identity)**.** *Let $\lambda \in \mathbb{R}^d$ and let σ be any pure profile with $\sigma_i \in O_i(\lambda)$ for all i. Put $e = e(\lambda, \sigma)$.*

(a) $\Phi(\sigma) - G(\lambda) = \frac{1}{2}\|e\|_M^2$. *Hence* $\Phi(\sigma) - \min_{\sigma'} \Phi(\sigma') \leq \Phi(\sigma) - \bar{\Phi}^* \leq \frac{1}{2}\|e\|_M^2$ *and* $G(\lambda^*) - G(\lambda) \leq \frac{1}{2}\|e\|_M^2$.

(b) For every carrier i, the Nash regret satisfies $J_i(\sigma_i; \sigma_{-i}) - \min_p J_i(p; \sigma_{-i}) \leq \frac{1}{2}\|e\|_{M,K_i}^2$.

(c) Let $m_i = M^{-1}\lambda - a_i(\sigma_i) = A_{-i}(\sigma) + e$. Then $\sigma_i \in \arg\min_p \{c_i(p) + \langle a_i(p), M m_i \rangle\}$: σ_i is an exact best response to the belief m_i about the rivals' aggregate. Consequently, if $m_{ik} \geq 0$ for $k \in K_i$, then σ_i is E-admissible (hence maximal and interval-undominated) for every credal box of half-width $\delta \geq \max_{k \in K_i} |e_k|/w_k$ centred at $A_{-i}(\sigma)$, in the sense of `main.tex`*, Definition 3.2 and Lemma 3.3.*

(d) If $e = 0$, then σ minimises Φ globally, it is a pure Nash equilibrium, $\lambda = \lambda^*$, and the convexification has no gap: $\min \Phi = \bar{\Phi}^*$.

(e) Conversely, if some pure profile σ° satisfies $\Phi(\sigma^\circ) = \bar{\Phi}^*$, then $\lambda^* = MA(\sigma^\circ)$ and $\sigma_i^\circ \in O_i(\lambda^*)$ for all i. In words, exact market clearing is attainable if and only if the convexification of the potential is tight at a pure profile.

Proof. First an identity valid for *every* pure σ and every λ. Let $A = A(\sigma)$ and $e = M^{-1}\lambda - A$, so that $\lambda = M(A + e)$. Then

$$\begin{aligned} \sum_i \left[\tilde{c}_i(\sigma_i) + \langle \lambda, a_i(\sigma_i) \rangle\right] - \tfrac{1}{2}\|\lambda\|_{M^{-1}}^2 &= \sum_i \tilde{c}_i(\sigma_i) + \langle M(A+e), A \rangle - \tfrac{1}{2}\|A + e\|_M^2 \\ &= \sum_i \tilde{c}_i(\sigma_i) + \tfrac{1}{2}\|A\|_M^2 - \tfrac{1}{2}\|e\|_M^2 = \Phi(\sigma) - \tfrac{1}{2}\|e\|_M^2. \qquad (*) \end{aligned}$$

(a) If $\sigma_i \in O_i(\lambda)$, the left side of $(*)$ equals $G(\lambda)$, which gives the identity. By Lemma 1(a), $G(\lambda) \leq G(\lambda^*) = \bar{\Phi}^* \leq \min \Phi$. So $\Phi(\sigma) - \min \Phi \leq \Phi(\sigma) - \bar{\Phi}^* \leq \Phi(\sigma) - G(\lambda) = \frac{1}{2}\|e\|_M^2$. Likewise $G(\lambda^*) - G(\lambda) = \bar{\Phi}^* - G(\lambda) \leq \Phi(\sigma) - G(\lambda)$.

(b) Fix i, write $a = a_i(\sigma_i)$ and, for a plan p, $a' = a_i(p)$ and $\Delta = a' - a$. Then $\Delta_k = 0$ for $k \notin K_i$. Since $MA_{-i} = \lambda - M(a+e)$,

$$\begin{aligned} J_i(p) - J_i(\sigma_i) &= c_i(p) - c_i(\sigma_i) + \langle \Delta, \lambda - Ma - Me \rangle \\ &= \big[\tilde{c}_i(p) + \langle \lambda, a' \rangle\big] - \big[\tilde{c}_i(\sigma_i) + \langle \lambda, a \rangle\big] + \tfrac{1}{2}\|a'\|_M^2 - \tfrac{1}{2}\|a\|_M^2 - \langle \Delta, Ma \rangle - \langle \Delta, Me \rangle \\ &\geq \tfrac{1}{2}\|\Delta\|_M^2 - \langle \Delta, Me \rangle \;\geq\; -\tfrac{1}{2}\|e\|_{M,K_i}^2. \end{aligned}$$

The bracket difference is ≥ 0 because $\sigma_i \in O_i(\lambda)$. We used $\frac{1}{2}\|a'\|^2 - \frac{1}{2}\|a\|^2 - \langle a' - a, Ma \rangle = \frac{1}{2}\|\Delta\|_M^2$, and the last step minimises $\frac{1}{2}\sum_{k \in K_i} M_k \Delta_k^2 - \sum_{k \in K_i} M_k \Delta_k e_k$ coordinatewise, at $\Delta_k = e_k$.

(c) For any p, with a, a' as in (b),

$$c_i(p) + \langle a', Mm_i \rangle = \big[\tilde{c}_i(p) + \langle \lambda, a' \rangle\big] + \tfrac{1}{2}\|a'\|_M^2 - \langle a', Ma \rangle = \big[\tilde{c}_i(p) + \langle \lambda, a' \rangle\big] + \tfrac{1}{2}\|a' - a\|_M^2 - \tfrac{1}{2}\|a\|_M^2.$$

The bracket is minimised by σ_i, and the middle term is ≥ 0 with equality at $p = \sigma_i$. Hence σ_i minimises the left side. For the credal statement, $J_i(p; m)$ depends on m only through the coordinates in K_i up to a constant independent of p, because $a_{ik}(p)$ is constant in p for $k \notin K_i$. So we may replace m_i by $\tilde{m}_i$ with $\tilde{m}_{ik} = m_{ik}$ for $k \in K_i$ and $\tilde{m}_{ik} = A_{-i,k}(\sigma)$ otherwise, without changing the argmin. Then $\tilde{m}_i \geq 0$ and $|\tilde{m}_{ik} - A_{-i,k}| = |e_k| \leq \delta w_k$, so $\tilde{m}_i$ lies in the clipped box $K^\delta(A_{-i}(\sigma))$. Under D1 the exposure equals the contribution, so by Lemma 3.3 of `main.tex` the Dirac law at $(\tilde{m}_i, \hat{\xi})$ is in the credal set (traffic is held at $\hat{\xi}$, which enters c_i as a constant price), and σ_i is optimal under it. That is E-admissibility, and Lemma 3.4 of `main.tex` gives maximality and interval dominance.

(d) If $e = 0$, (a) gives $\Phi(\sigma) = G(\lambda) \leq \bar{\Phi}^* \leq \min \Phi \leq \Phi(\sigma)$, so all are equal. A global minimiser of an exact potential is a pure Nash equilibrium. $G(\lambda) = \max G$ and uniqueness give $\lambda = \lambda^*$.

(e) The vertex of X given by σ° minimises $\bar{\Phi}$. By Lemma 1(b), $\lambda^* = MA(\sigma^\circ)$, so $e(\lambda^*, \sigma^\circ) = 0$. By $(*)$, $\sum_i [\tilde{c}_i(\sigma_i^\circ) + \langle \lambda^*, a_i(\sigma_i^\circ) \rangle] - \frac{1}{2}\|\lambda^*\|_{M^{-1}}^2 = \Phi(\sigma^\circ) = \bar{\Phi}^* = G(\lambda^*)$. Subtracting $G(\lambda^*)$ gives $\sum_i \{\tilde{c}_i(\sigma_i^\circ) + \langle \lambda^*, a_i(\sigma_i^\circ) \rangle - V_i(\lambda^*)\} = 0$, a sum of nonnegative terms. So each term is 0, that is, $\sigma_i^\circ \in O_i(\lambda^*)$. □

Remark 3 (What is new in Theorem 2). Identity $(*)$ is Fenchel–Young equality for $\frac{1}{2}\|\cdot\|_M^2$, and (a) is an instance of the familiar fact that a Lagrangian dual gap at a primal-feasible point measures suboptimality. The content specific to the game is in (b) and (c). The Nash regret and the credal width are controlled by the *total* clearing error e. The own-contribution term $a_i(\sigma_i) - \bar{a}_i$, which appears in the absorption bound of `main.tex` (Theorem 6.1(b); Remark 6.3(2) shows it cannot be removed there) and in the $(\sqrt{q} + 1)$ of Liu–Oudjane–Wan, cancels exactly. The anchor $m_i = M^{-1}\lambda - a_i(\sigma_i)$ depends on the carrier's own plan. This is possible only because, under D1, the own quadratic term turns a pricing-optimal plan into a best response to that anchor (proof of (c)). The bound in (b) is local: carrier i is affected only by clearing errors in its own zones.

Theorem 2 turns the search for a good pure equilibrium into a *price search*: find one common λ at which the carriers' pricing answers nearly clear every zone. The next section shows that this search can be done one zone at a time, with all zones in parallel.

3 Diagonal monotonicity and a dimension-free bracket

Definition 4 (Cross-zone variation)**.** For $k \in [d]$ let

$$\kappa_k = \sup\Big\{ \sup_{r \in R(\lambda)} \inf_{r' \in R(\lambda')} |r_k - r'_k| \;:\; \lambda, \lambda' \in B,\ \lambda_k = \lambda'_k \Big\}.$$

κ_k measures how far the total response in coordinate k can be moved by the prices of the *other* coordinates at a fixed own price. Always $\kappa_k \le A_k^{\max}$.

Proposition 5 (Separable plan sets)**.** *If every $\mathcal{P}_i$ is a product $\prod_k \mathcal{P}_{ik}$, with $a_{ik}(p) = \alpha_{ik}(p_k)$ and $\tilde{c}_i(p) = \sum_k \gamma_{ik}(p_k)$, then $\kappa_k = 0$ for all k.*

Proof. $V_i(\lambda) = \sum_k v_{ik}(\lambda_k)$ with $v_{ik}(x) = \min_{p_k}\{\gamma_{ik}(p_k) + x\alpha_{ik}(p_k)\}$. The superdifferential of a separable concave function is the product of the one-dimensional superdifferentials, and $R(\lambda) = \sum_i \partial^+ V_i(\lambda)$ is again a product. Its projection on coordinate k is $\sum_i \partial^+ v_{ik}(\lambda_k)$, which depends on λ_k only. For $\lambda_k = \lambda'_k$ the projections coincide, so every inner infimum is 0. □

In the routing application, separability holds when every vehicle stays in one zone and zone fleets are independent, and the coordinates are deliveries only. Fleet sharing across zones, routes crossing zones, and the coupling between deliveries and vehicle-hours in a zone all make $\kappa_k > 0$. Example 9 shows that the floor below is then genuine.

Theorem 6 (Bracket invariant)**.** *Start from $l = 0$ and $u = (M_k A_k^{\max})_k$. In each round, query any $\lambda \in \prod_k [l_k, u_k]$, obtain oracle answers $\sigma_i \in O_i(\lambda)$, and set $s = A(\sigma) - M^{-1}\lambda$ (which equals $-e$). For each k: if $s_k > 0$ set $l_k \leftarrow \lambda_k$; if $s_k < 0$ set $u_k \leftarrow \lambda_k$; if $s_k = 0$ set $l_k = u_k \leftarrow \lambda_k$. Then, after every round and for every k,*

$$\operatorname{dist}\left(\lambda_k^*, [l_k, u_k]\right) \le M_k \kappa_k.$$

Proof. Key inequality. Let $\lambda \in B$ be a query with $\lambda_k = \mu$ and observed $r = A(\sigma) \in R(\lambda)$. We claim:

(i) $s_k > 0 \Rightarrow \lambda_k^* \ge \mu - M_k\kappa_k$; (ii) $s_k < 0 \Rightarrow \lambda_k^* \le \mu + M_k\kappa_k$; (iii) $s_k = 0 \Rightarrow |\lambda_k^* - \mu| \le M_k\kappa_k$.

For (i), suppose $\mu - \lambda_k^* > M_k\kappa_k$. Let λ' agree with λ^* except $\lambda'_k = \mu$. Then $\lambda' \in B$, because $\lambda^* \in B$ (Lemma 1(b)) and $\mu \in [0, M_k A_k^{\max}]$. By Lemma 1(c) pick $r^* \in R(\lambda^*)$ with $r^* = M^{-1}\lambda^*$. Since V is concave, R is antitone: $\langle r' - r^*, \lambda' - \lambda^* \rangle \le 0$ for every $r' \in R(\lambda')$. Here $\lambda' - \lambda^* = (\mu - \lambda_k^*)e_k$ with $\mu - \lambda_k^* > 0$, so $r'_k \le r_k^* = \lambda_k^*/M_k$ for every $r' \in R(\lambda')$. Because $\lambda_k = \lambda'_k$, the definition of κ_k gives, for every $\eta > 0$, some $r' \in R(\lambda')$ with $r_k \le r'_k + \kappa_k + \eta$. Hence

$$s_k = r_k - \mu/M_k \le \lambda_k^*/M_k + \kappa_k + \eta - \mu/M_k = -(\mu - \lambda_k^* - M_k\kappa_k)/M_k + \eta.$$

This is negative for small η, contradicting $s_k > 0$. Statement (ii) is symmetric. For (iii), if $\mu - \lambda_k^* > M_k\kappa_k$ the same computation gives $s_k < 0$, and if $\lambda_k^* - \mu > M_k\kappa_k$ the symmetric one gives $s_k > 0$.

Induction. Initially $\lambda_k^* \in [0, M_k A_k^{\max}]$, so the distance is 0. Suppose $\operatorname{dist}(\lambda_k^*, [l_k, u_k]) \le M_k\kappa_k$ and a round queries $\mu = \lambda_k \in [l_k, u_k]$. If $s_k > 0$, the new bracket is $[\mu, u_k]$. If $\lambda_k^* \ge \mu$, its distance to $[\mu, u_k]$ is $\max(0, \lambda_k^* - u_k)$, which is the old distance. If $\lambda_k^* < \mu$, the distance is $\mu - \lambda_k^* \le M_k\kappa_k$ by (i). The case $s_k < 0$ is symmetric with (ii). If $s_k = 0$, the new bracket is $\{\mu\}$ and (iii) applies. □

Corollary 7 (Dimension-free oracle bound)**.** *Run Theorem 6 for T rounds and query the bracket midpoint at least every other round (other rounds may use any point of the bracket, for example a secant step). Then every point λ of the final bracket satisfies, for all k,*

$$|\lambda_k - \lambda_k^*| \le M_k A_k^{\max} 2^{-\lceil T/2 \rceil} + M_k\kappa_k,$$

and

$$G(\lambda^*) - G(\lambda) \le 2\sum_k M_k (A_k^{\max})^2 2^{-\lceil T/2 \rceil} + 2\sum_k M_k A_k^{\max}\kappa_k.$$

The number of oracle calls is NT, independent of d. The N calls of a round are independent and can run in parallel. With $\kappa = 0$ (Proposition 5), $\lambda \to \lambda^*$ geometrically in every coordinate.

Proof. Midpoint rounds halve the width of every bracket, or reduce it to a point. Other rounds never increase it, since the new bracket is a sub-interval. With midpoints in at least $\lceil T/2 \rceil$ of the rounds, the final width is at most $M_k A_k^{\max} 2^{-\lceil T/2 \rceil}$. Adding the distance bound of Theorem 6 gives the first claim. For the second, take any $g \in \partial^+ G(\lambda) = R(\lambda) - M^{-1}\lambda$. Concavity gives $G(\lambda^*) - G(\lambda) \le \langle g, \lambda^* - \lambda \rangle \le \sum_k |g_k||\lambda_k^* - \lambda_k|$. Since $0 \le r_k \le A_k^{\max}$ and $0 \le \lambda_k / M_k \le A_k^{\max}$ on B, $|g_k| \le A_k^{\max}$; the factor 2 is a safe rounding of this bound. Substitute the first claim. □

Remark 8 (Why this beats black-box methods on the same dual). G is strongly concave but non-smooth, and it is available only through the oracle. Frank–Wolfe on $\bar{\Phi}$ (`main.tex`, Theorem 8.4) and the subgradient method need $\Theta(1/\varepsilon)$ rounds in the worst case. Cutting-plane and column-generation methods (Kelley, simplicial decomposition, bundle) need a number of rounds that grows with d: a cutting-plane model is accurate only after about d cuts. Corollary 7 exploits two facts that a black-box method ignores. The coupling is diagonal (one supply line per coordinate), and every oracle answer reveals a full d-dimensional supergradient, which is *one sample for each of the d one-dimensional root-finding problems simultaneously.* The price of this is the floor $M_k \kappa_k$.

Example 9 (The floor is genuine)**.** Let $d = 2$, $M = I$ and one carrier with two plans: 0 ($\tilde{c} = 0$, $a = (0,0)$) and a bundle b ($\tilde{c} = -1.5$, $a = (1,1)$), for instance one vehicle that must visit both zones. Then $A^{\max} = (1,1)$, $B = [0,1]^2$ and $V(\lambda) = \min\{0, -1.5 + \lambda_1 + \lambda_2\}$. The maximiser of G lies on the kink line $\lambda_1 + \lambda_2 = 1.5$ (off the line, G is $-\frac{1}{2}|\lambda|^2$ or $-1.5 + \lambda_1 + \lambda_2 - \frac{1}{2}|\lambda|^2$, whose unconstrained maximisers $(0,0)$ and $(1,1)$ lie on the wrong side). Hence $\lambda^* = (0.75, 0.75)$, with $R(\lambda^*) = \mathrm{conv}\{(0,0),(1,1)\} \ni \lambda^*$. For own price $\lambda_1 = x \in [0.5, 1]$ the response in coordinate 1 is 1 or 0 depending on whether $\lambda_2 < 1.5 - x$, so $\kappa_1 = \kappa_2 = 1$. The queries $\lambda = (0.7, 0.9)$, $(0.6, 1)$ and $(0.5, 1)$ (with the tie at the last one resolved towards plan 0) all return plan 0, so $s_1 = -\lambda_1 < 0$ and the bracket update sets $u_1 = 0.5$. The bracket then excludes $\lambda_1^* = 0.75$ by 0.25, which is within the floor $M_1 \kappa_1 = 1$ of Theorem 6 but positive. With complementary coordinates, the sign of one zone's residual depends on the other zone's price, and no rule that looks at one coordinate at a time can avoid this.

4 The algorithm

The secant step implements the learning idea of direction (a) in its simplest provable form. Each coordinate's total response is a nonincreasing function of its own price up to the cross-variation, and the isotonic fit keeps a monotone surrogate of it, built from all past oracle answers at once. A full concave surrogate of V_i (a cutting-plane model) would be column generation; we deliberately keep only the d one-dimensional monotone surrogates, because that is what makes the round count independent of d.

Theorem 10 (Guarantee of ZMC)**.** *Under Assumptions 1–2, with exact oracles:*

(a) ZMC uses NT pricing calls before the polish, and the polish stops after finitely many best-response calls with a pure Nash equilibrium σ.

(b) $\Phi(\sigma) \le \Phi(\sigma^{\mathrm{price}}) = G(\lambda^{t^\star}) + \frac{1}{2}\|e^{t^\star}\|_M^2 \le \bar{\Phi}^* + \frac{1}{2}\min_t \|e^t\|_M^2$.

(c) If the minimal positive potential improvement of a unilateral deviation is $\nu > 0$, the polish makes at most $\frac{1}{2}\min_t \|e^t\|_M^2 / \nu$ plan changes, and hence at most $N(1 + \frac{1}{2}\min_t \|e^t\|_M^2/\nu)$ best-response calls.

Algorithm 1 ZMC: Zone-Market Clearing

Require: pricing oracles O_i (Assumption 2); best-response oracles; rounds T

$l \leftarrow 0$, $u \leftarrow MA^{\max}$, $\lambda \leftarrow (l+u)/2$; pools $\mathcal{Q}_i \leftarrow \emptyset$; samples $\mathcal{S} \leftarrow \emptyset$

for $t = 0, \dots, T-1$ **do**

  for all i **in parallel**: $\sigma_i^t \leftarrow O_i(\lambda)$; $\mathcal{Q}_i \leftarrow \mathcal{Q}_i \cup \{\sigma_i^t\}$ ▷ N routing solves

  $A^t \leftarrow \sum_i a_i(\sigma_i^t)$, $s \leftarrow A^t - M^{-1}\lambda$, $G^t \leftarrow \sum_i[\tilde{c}_i(\sigma_i^t) + \langle \lambda, a_i(\sigma_i^t)\rangle] - \frac{1}{2}\|\lambda\|^2_{M^{-1}}$

  update every bracket $[l_k, u_k]$ by the sign of s_k (Theorem 6); $\mathcal{S} \leftarrow \mathcal{S} \cup \{(\lambda, A^t)\}$

  if t odd **then** $\lambda \leftarrow (l+u)/2$

  else for each k: fit a nonincreasing (isotonic) piecewise-linear $\hat{r}_k$ to the pairs $(\lambda_k, A_k) \in \mathcal{S}$;

    $\lambda_k \leftarrow$ the crossing of $\hat{r}_k$ with $x \mapsto x/M_k$, clipped to $[l_k, u_k]$ ▷ secant step

$\sigma^{\text{price}} \leftarrow \sigma^{t^\star}$ with $t^\star = \arg\min_t \frac{1}{2}\|e^t\|^2_M$ (equivalently, by Theorem 2(a), the best $\Phi(\sigma^t)$)

$x \leftarrow \arg\min\{\bar{\Phi}(x) : x_i \in \Delta(\mathcal{Q}_i)\}$ (restricted master, no oracle); $\sigma^{\text{round}} \leftarrow$ derandomised rounding of x (`main.tex`, eq. (8.1))

$\sigma \leftarrow \arg\min\{\Phi(\sigma^{\text{price}}), \Phi(\sigma^{\text{round}})\}$; run sequential best responses from σ until no carrier improves

return σ (a pure Nash equilibrium; optionally the add-on certificate of Section 6)

(d) σ is an exact pure equilibrium, so $\sigma \in \mathrm{ICNE}_C(\delta)$ for all $\delta \geq 0$ and $C \in \{E, M, ID\}$. If ZMC is run on the surcharged game of `main.tex`, Lemma 3.6, σ is an exact Γ-maximin equilibrium at width δ.

Proof. (a) Each round makes one pricing call per carrier. The polish is sequential best-response dynamics in a finite exact-potential game. Every accepted change strictly decreases Φ, which takes finitely many values, so it stops. At the stop, no carrier has a strictly improving deviation, which is a pure Nash equilibrium. (b) The polish does not increase Φ, and the start has $\Phi \leq \Phi(\sigma^{\text{price}})$. The equality is Theorem 2(a) applied to $\lambda^{t^\star}$ and $\sigma^{t^\star}$, and $G \leq \bar{\Phi}^*$ by Lemma 1(a). (c) Each change decreases Φ by at least ν, and Φ starts at most $\frac{1}{2}\min_t \|e^t\|^2_M$ above $\bar{\Phi}^* \leq \min \Phi$. Between changes, at most N calls are made without a change, and N more certify the end. (d) An exact pure Nash equilibrium makes each plan optimal at the centre belief, which lies in every credal box (`main.tex`, Proposition 7.5). The Γ case is Lemma 3.6 of `main.tex`. □

Remark 11 (Heuristic oracles). With a heuristic routing oracle, (b) still holds with G replaced by the computed value, which is then only an upper estimate of V. The equilibrium claim of (a) becomes "no improvement found by the oracle". A lower bound $\mathcal{U}_i$ on V_i (column-generation LP) restores a valid bound as in `main.tex`, Theorem 8.5(c).

5 Charges: warm-started sweeps for the leader

With charges $t \in \mathbb{R}^d$ on the contributions, $\tilde{c}_i$ becomes $\tilde{c}_i + \langle t, a_i\rangle$, so $V^t(\lambda) = V(\lambda + t)$. With the substitution $\mu = \lambda + t$, the equilibrium condition (Lemma 1(c)) reads $M^{-1}(\mu - t) \in R(\mu)$, where R does not depend on t. Charges translate the supply line.

Proposition 12 (Lipschitz dependence on charges)**.** *Let $\mu^*(t) = \lambda^*(t) + t$ be the equilibrium gross price under charges t. Then $\|\mu^*(t) - \mu^*(t')\|_{M^{-1}} \leq \|t - t'\|_{M^{-1}}$, and the convexified equilibrium aggregate $\bar{A}^*(t) = M^{-1}(\mu^*(t) - t)$ satisfies $\langle \bar{A}^*(t) - \bar{A}^*(t'), t - t'\rangle \leq 0$. That is, aggregates respond monotonically to charges.*

Proof. Let $r = M^{-1}(\mu - t) \in R(\mu)$ and $r' = M^{-1}(\mu' - t') \in R(\mu')$ with $\mu = \mu^*(t)$ and $\mu' = \mu^*(t')$. Antitonicity gives $\langle r - r', \mu - \mu' \rangle \leq 0$, that is, $\|\mu - \mu'\|^2_{M^{-1}} \leq \langle M^{-1}(t - t'), \mu - \mu' \rangle \leq \|t - t'\|_{M^{-1}} \|\mu - \mu'\|_{M^{-1}}$ by Cauchy–Schwarz in the M^{-1} inner product. For the second claim, $\langle r - r', t - t' \rangle = \langle r - r', (\mu - \mu') - M(r - r') \rangle \leq -\|r - r'\|^2_M \leq 0$. □

Consequently, when the leader evaluates a sequence of charge vectors, ZMC for t' can start from the bracket $\mu^*_k(t) \pm \sqrt{M_k}\, \|t - t'\|_{M^{-1}}$ (then $\lambda = \mu - t'$), which by Corollary 7 saves about $\log_2(M_k A^{\max}_k / (\sqrt{M_k} \|t - t'\|_{M^{-1}}))$ rounds per evaluation. The plan pools $\mathcal{Q}_i$ are reusable across charges, because the plan sets do not depend on t. We do not claim an algorithm for the optimal charge (a bilevel problem with an NP-hard follower); the proposition only makes each follower solve cheaper.

6 Optional add-on: a posteriori quality statement

The quantities computed during ZMC give, at no extra oracle cost, the bracket $\bar{\Phi}^* \in [\max_t G^t,\ \Phi(\sigma)]$ when the oracle is exact (Lemma 1, Theorem 2(a)). For each pricing profile σ^t, the clearing identity also gives explicit Nash regrets $\leq \frac{1}{2}\|e^t\|^2_{M,K_i}$ and credal widths $\max_{k \in K_i} |e^t_k| / w_k$ (Theorem 2(b),(c)). This is a by-product; the output of ZMC is the equilibrium σ.

7 Limits and relation to prior art

Limits. (1) The floor $M_k \kappa_k$ is real (Example 9). In the prototypes the relative dual gap of ZMC stalls near 10^{-3} while column generation goes to 0. This floor is below the nonconvexity (rounding) gap of the pure problem, so for pure equilibria it did not matter in our tests. It would matter if one needed the convexified optimum itself to high accuracy. (2) The method needs the potential structure: D1 with symmetric diagonal M, or Γ-maximin via the surcharge. Under D2 there is no common price at which pricing-optimality implies best response, and Theorem 2(b),(c) fail. We have no D2 version. (3) Theorem 10 does not bound $\min_t \|e^t\|_M$ a priori. Corollary 7 controls the *price*, not the clearing error of pure answers. At λ^*, carriers whose response is at a kink must mix, and the irreducible error is the Shapley–Folkman/rounding scale of `main.tex`. (4) The city experiments use a zone-level routing proxy with an exact dynamic-programming oracle (one fleet constraint per carrier), not a street-level VRP.

Prior art (searches of 2026-10-23; see `algorithm.md`). Lagrangian or price decomposition with a pricing oracle is classical (Dantzig–Wolfe, column generation, bundle methods; Briant et al., Math. Program. 2008, compare bundle and column generation on vehicle routing). Diagonal variable-metric bundle methods exist (Karmitsa, Gaudioso and Joki, 2019). Learning separable piecewise-linear concave value functions from sampled slopes is SPAR (Powell, Ruszczyński and Topaloglu, Math. Oper. Res. 2004). Walrasian prices with indivisible goods from aggregate demand oracles are computed by convex minimisation of a market potential (Paes Leme and Wong, 2017, $\tilde{O}(n)$ queries under gross substitutes, via cutting planes). Coordinate tâtonnement is studied for Fisher markets. Duguet, Harks, Schmidt and Schwarz (2025) relate the regret of price-allocation pairs to duality and integrality gaps in combinatorial markets. What we did not find is the combination used here: the own-term cancellation that makes a common price yield best responses to anchors in a potential aggregative game (Theorem 2(b),(c),(e)), the sign-bracket invariant with the cross-variation floor (Theorem 6), and its use to obtain d-independent round counts for pure equilibria with an NP-hard oracle. The individual tools (bisection, secant, isotonic regression, rounding,

best-response polish) are standard, and we claim novelty only for the structural theorems and for the method they define.